# Comparative Efficacy Of Fixed And Adaptive Torque Modulation In Low-Cost Myoelectric Prosthetic Grasping

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## Abstract

EMG-controlled upper-limb prosthetics typically rely on fixed-threshold actuation, applying maximum torque whenever a signal crosses a set threshold, which causes mechanical failure on compliant objects and excludes users with low-amplitude EMG signals. This study tested whether proportional EMG-to-torque mapping outperforms binary threshold actuation on a \$118.45 consumer platform (PandaWare: RP2040 microcontroller, MyoWare 2.0 EMG sensor, MG996R servo, INA219 current monitor) across standardized grasping trials ($n = 12$ per condition) using deformable, frictional, and high-mass objects. Adaptive Mode preserved fragile objects 100% of the time versus 40% in Fixed Mode, using closed-loop stall detection that arrests PWM once current exceeds 450 mA, compared to uncontrolled spikes reaching 1800 mA in Fixed Mode. Phase-specific analysis showed the system operates in two distinct phases, proportional closure followed by arrest-protocol current suppression, which explains why the aggregate correlation is near zero (Fixed $r = 0.29$; Adaptive $r = -0.06$). Step-normalized energy consumption was also 14.0% lower in Adaptive Mode (10.48 vs. 12.19 mJ/step; $t = -2.70$, $p = 0.007$), and Neuro-Adaptive Normalization eliminated signal-to-action failure by computing personalized EMG amplitude ranges for each user. Together, these results show that high-fidelity adaptive prosthetic control can be achieved through software alone, without specialized hardware.

## Keywords

*Biomedical Engineering, Biomedical Devices, Electromyography, Proportional Control, Haptic Feedback*

## INTRODUCTION

### The Prosthetic Accessibility Gap

Upper-limb amputation affects an estimated 57.7 million people worldwide, the majority of whom reside in low- and middle-income countries with limited access to rehabilitation technology (1). Advanced bionic hands typically range from \$5,000 to \$50,000 USD (2). This places them beyond the financial reach of approximately 90% of amputees globally (3). Consequently, most patients are left with passive cosmetic devices or no prosthetic device at all, substantially limiting their occupational and social participation. This study proceeds from the premise that the performance gap between hobbyist-grade and clinical-grade prosthetics is largely attributable to software rather than hardware, and that this gap can therefore be closed at minimal cost.

Affordability analysis supports this premise. The PandaWare prototype was constructed for \$118.45 USD using off-the-shelf retail components and 3D printing. Transitioning to a custom PCB with a surface-mount AD8232 instrumentation amplifier would reduce the Bill of Materials to approximately $19.22 (4). After accounting for current Section 301 import tariffs (25%) and a 35% margin covering assembly, local taxes, and reinvestment, the projected end-user price comes to $49.49, a 97.5% reduction relative to entry-level commercial myoelectric devices (5).

### Limitations of Fixed-Actuation Control

Standard myoelectric prosthetics use fixed-actuation logic: a static voltage threshold is applied to the rectified EMG envelope, and any signal that crosses this threshold commands the motor to spin to maximum extension at full torque. While straightforward to implement, this approach introduces three fundamental failures. First, it is physiologically exclusionary: users with muscular atrophy, pediatric users, or individuals with characteristically low-amplitude EMG signals may never exceed the static threshold, resulting in complete signal-to-action failure: the motor simply does not turn. Second, it is mechanically destructive for compliant objects: because maximum torque is applied regardless of object properties, fragile items such as paper cups, soft packaging, or medical containers are likely to be deformed or crushed. Third, it is energetically wasteful: the motor continuously draws saturation current regardless of whether additional grip force is required, which accelerates battery depletion and causes the motor to overheat (6).

Research has shown that control latency exceeding 100 ms degrades a user's sense of prosthetic agency and increases cognitive load during manipulation tasks (7). Moreover, most low-cost systems lack shape synergy, which is the ability to modulate grip conformation and force based on the object properties encountered during a grasp (8). Without this capability, users often have to compensate manually, adjusting their approach or applying additional muscular effort to account for what the prosthetic itself fails to sense. This limitation can be addressed through proportional control architecture without requiring any additional hardware costs.

### Adaptive Torque Control

Adaptive torque control replaces the binary threshold with continuous proportional mapping: normalized EMG amplitude is translated directly into servo angle, allowing the user to control grip force through the intensity of their muscle contraction rather than a binary on-off response. EMG-based torque estimation has previously been demonstrated in research-grade exoskeletons (9). However, these implementations rely on expensive clinical-grade sensors, dedicated DSP hardware, or invasive electrodes for signal acquisition. This study tests whether comparable performance can be achieved on a $118.45 budget through algorithmic and mechanical design alone.

### Objectives and Hypotheses

This study addresses two primary hypotheses. The first, $\mathbf{H_1}$, was that proportional EMG-to-torque mapping would outperform fixed-threshold actuation across all three failure modes. The second, $\mathbf{H_2}$, was that users would experience higher signal-to-action failure rates under fixed-threshold actuation than under personalized amplitude normalization.

## METHODS AND MATERIALS

### Hardware Architecture

The PandaWare prosthetic system was built around a Raspberry Pi Pico (RP2040) dual-core ARM Cortex-M0+ microcontroller, chosen for its 12-bit ADC, native I²C bus support, 133 MHz clock speed enabling sub-millisecond processing loops, and retail cost under $5.10 (10). EMG acquisition was performed using a MyoWare 2.0 Muscle Sensor with snap-on foam bipolar surface electrodes placed over the flexor digitorum superficialis, with an inter-electrode distance of 20 mm consistent with SENIAM surface EMG placement guidelines (11). The MyoWare 2.0 performs on-chip full-wave rectification and

RC low-pass filtering, which delivers a smoothed EMG envelope (ENV) signal to the ADC at electrical speeds and offloads the computationally expensive signal cleaning from the microcontroller (12).

Actuation was provided by a single MG996R high-torque servo (stall torque: 11 kg·cm at 6V), controlled through a PCA9685 16-channel PWM driver at 12-bit resolution over $I^2C$. The chassis was 3D-printed in PLA on an Elegoo Centauri Carbon FDA printer. The tendon-driven finger mechanism used 0.8 mm Dyneema cordage for active flexion, with a servo horn turn radius of approximately 0.75 inches, and 1.5 mm elastic shock cord for passive extension return. A 65° multi-axial thumb joint enabled anatomically accurate opposition grip. Power was supplied by a 7.4V 2S LiPo battery regulated to 5.0V via a DC-DC buck converter, with a 10A physical fuse and software-level torque limits providing overcurrent protection. Real-time current monitoring was provided by an INA219 bidirectional current/power monitor (Texas Instruments) (13).

A critical architectural decision was the implementation of a Dual-Bus $I^2C$ topology. The primary bus ($I^2C0$) was dedicated exclusively to the PCA9685 PWM driver for servo positioning. The secondary bus ($I^2C1$) polled the INA219 at 50 Hz independently. This hardware isolation prevented high-frequency current sensor interrupts from inducing PWM jitter, maintaining a biomimetic motor response latency below 20 ms. Total production cost amounted to $118.45 USD.

**Signal Processing Pipeline**

Firmware was implemented in MicroPython on Thonny IDE v4.1.4 (14). The processing pipeline operated at 1000 Hz to satisfy the Nyquist-Shannon criterion for the 20–500 Hz muscle fiber firing range, preventing aliasing of motor unit action potentials (15, 16). Six sequential stages constituted the pipeline:

***ADC Digitization:*** Raw MyoWare ENV signal sampled via GP26 at 12-bit resolution.

***Hardware Envelope Extraction:*** MyoWare 2.0 on-chip full-wave rectification $|V_{ra}{}^{W}|$ passed through RC low-pass network; capacitor integrated spike energy into smooth envelope. This saves CPU cycles and extends battery life (12).

***Software Averaging:*** ENV sampled 50 times per control loop and arithmetically averaged, eliminating Gaussian noise and ensuring a stable, reproducible baseline.

***Neuro-Adaptive Normalization:*** $S_{norm} = (S - S_{rest}) / (S_{max} - S_{rest})$. Personalized quiescent noise floor $S_{rest}$ and peak voluntary contraction $S_{max}$ are recorded during a pre-trial MVC calibration, mapping each user's idiosyncratic EMG amplitude range onto a universal 0–1 scale.

***Power Scaling:*** $S_{norm}$ raised to the 0.8 power: if $S_{norm}$ = 0.2, output = 0.2^0.8 = 0.27. This non-linear stretch ensures low-amplitude contractions produce proportional, not negligible, motor responses, preventing dead-band artifacts.

***Temporal Low-Pass Filter:*** Motor update frequency capped at 50 Hz, independent of the 1000 Hz ADC sampling rate. Changes faster than 50 Hz are suppressed, filtering high-frequency biological tremor and producing fluid servo motion.

**Control Conditions**

***Fixed Displacement Mode:*** A rigid 0.3V threshold applied to the raw ENV signal. Any threshold-breaking value immediately commands the servo to maximum extension (180°) at full torque. Motor output is binary: 0° or 180°, with no proportionality to EMG amplitude.

***Adaptive Displacement Mode:*** Normalized Snorm mapped proportionally to servo angle via the linear transfer function Imotor = G × Sinput − θ, where G is the calibration gain and θ is the quiescent offset. Simultaneously, the closed-loop haptic arrest protocol monitored INA219 current draw at 50 Hz; upon detecting I > 450 mA (indicating mechanical contact / stall), the PWM signal was immediately locked, arresting motor rotation and preventing object deformation.

**Experimental Procedure**

Trials were conducted during February 5–27, 2026 in Carson, California. Skin was prepared with 70% isopropyl alcohol to reduce electrode impedance before bipolar MyoWare 2.0 electrode placement over the flexor digitorum superficialis muscle belly. A pre-trial MVC calibration sequence recorded Srest and Smax for each subject. Three participants (n = 3) completed 12 trials per control condition in randomized order. This study involving human participants was formally reviewed and approved by the Los Angeles County Science and Engineering Fair (LACSEF) Scientific Review Committee (SRC) and Institutional Review Board (IRB) equivalent ethics committee, in strict accordance with the International Science and Engineering Fair (ISEF) guidelines. Signed informed consent was prospectively obtained from all adult participants, and there were no minors other than the researcher included in data collection. To preserve privacy and confidentiality, all collected electromyography (EMG) streams and biometric data were fully de-identified upon acquisition by stripping all personally identifiable information (PII) and indexing datasets using randomized, anonymous alphanumeric codes. Each trial followed the sequence: REACH → GRASP → LIFT → HOLD → RELEASE. A 90-second refractory period between trials mitigated biological muscle fatigue, consistent with established EMG fatigue recovery protocols (17). Three test objects were selected to probe distinct control system requirements:

***Object A:*** Deformable (Soft Paper Cup, ~10 g): A thin-walled cellulose standard chosen to isolate the closed-loop haptic arrest response. Success: secure grip without structural deformation.

***Object B:*** Frictional (Tennis Ball, ~56 g): A rigid curved-surface object chosen to probe the proportional control law under moderate load. Success: stable lift with minimal observable muscle effort.

***Object C:*** High Mass (Filled Water Bottle, ~500 g): A malleable high-inertia object chosen to test maximum torque scaling. Success: sustained lift against gravity without motor stall or object slip.

**Data Acquisition and Analysis**

Data was logged via the RP2040 to CSV at 1000 Hz. Fields per timestep: raw ADC value, ENV, normalized $S_{norm}$, final servo angle, INA219 current (mA), stall flag (0/1), and computed RMS over a trailing 50 ms window. Grasp outcome (success/failure/deformation) was recorded manually per trial. Post-hoc analysis included: (a) time-series overlay of ENV versus servo angle (Figure 1); (b) servo angle distribution histogram (Figure 2); (c) scatter plot of ENV versus current draw with phase-coded coloring (Figure 3); (d) phase-split correlation analysis (Figure 4); (e) time-series of current draw with stall flag overlay (Figure 5); (f) active grip duty cycle comparison (Figure 6); and (g) cumulative energy $E = \Sigma(V \times I \times \Delta t)$ calculated by discrete Riemann Sum across matched trial windows. Statistical significance for energy comparison established via two-sample t-test ($p < 0.05$ threshold). Quantitative time-series and correlation analyses reported in the Results and Discussion section were performed on Object A (deformable, n = 12 trials per condition), which was selected as the primary test case because it isolates all three control-law failure modes simultaneously. All hypotheses were formulated prior to data collection. The control law architectures were defined before experimental trials began, and outcome metrics were pre-specified to prevent post-hoc hypothesis modification.

## RESULTS AND DISCUSSION

### EMG Envelope vs. Servo Angle: Actuation Linearity

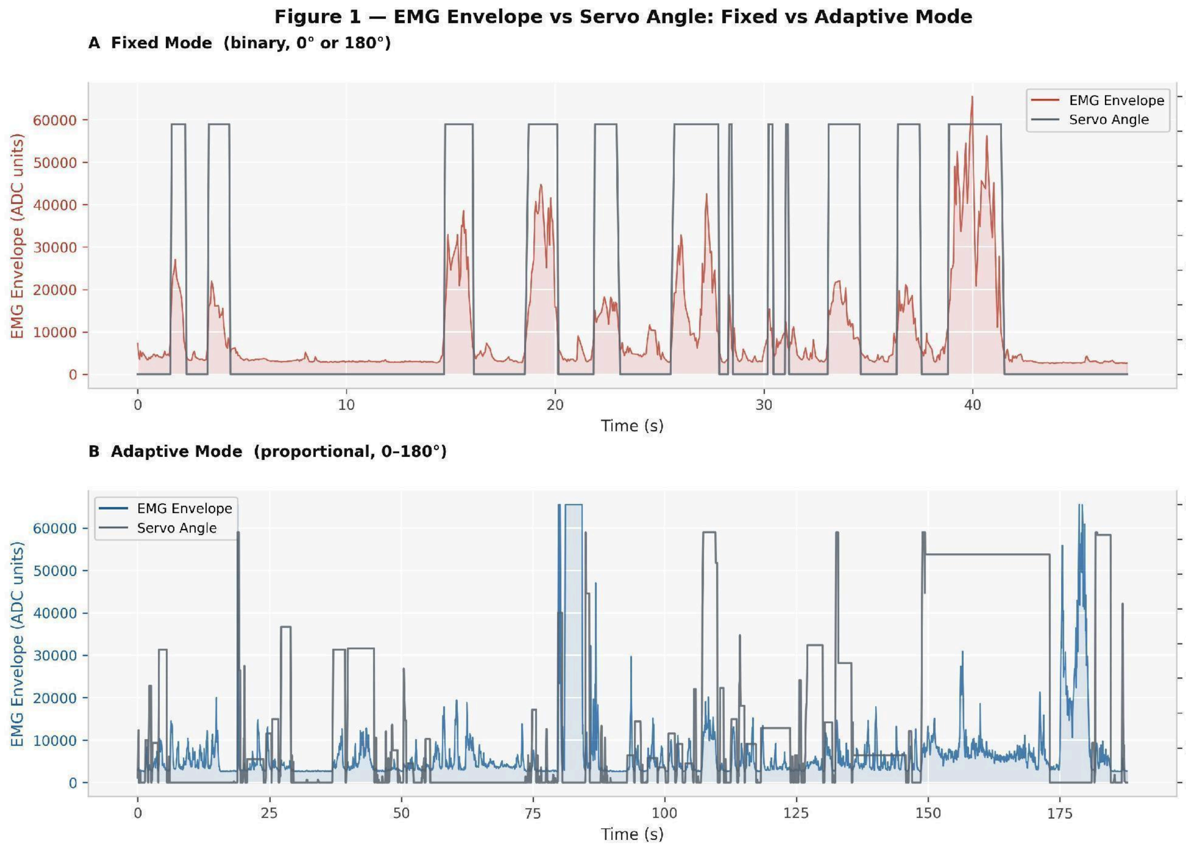


**Figure 1.** EMG envelope and servo angle versus time. Fixed Mode (A) shows binary square-wave actuation; Adaptive Mode (B) shows proportional tracking across the full 0°–180° range.

Figure 1 presents the primary behavioral characterization of both control modes over matched experimental sessions. In Fixed Mode (Figure 1A), the servo angle exhibits a characteristic square-wave profile: it jumps instantly to maximum extension (180°) upon threshold crossing and returns to 0° upon signal dropout, regardless of intermediate EMG amplitude variations. The EMG envelope shows natural biological variance: peaks of varying height corresponding to contractions of different intensities, but none of this variance is transmitted to the servo, which has a binary output. The motor responds identically to a 20% MVC contraction and a 100% MVC contraction. This binary relationship is the direct mechanical cause of fragile-object failure: the system delivers maximum torque even when the user's intent is clearly a gentle grip.

In Adaptive Mode (Figure 1B), the servo angle tracks the envelope with high fidelity across the full range of motion. Moderate contractions produce intermediate grip angles, and stronger contractions produce proportionally greater closure. The proportional mapping remains relatively stable across consecutive trials, confirming that Neuro-Adaptive Normalization successfully anchors the control law to each subject's physiology. The data directly supports $H_1$: the proportional control law is behaviorally verified at the servo-output level. Quantitative analysis of servo angle distribution confirms this structurally: Fixed Mode operated exclusively at binary positions, 70.9% of timesteps at 0° and 29.1% at 180° with zero intermediate states, while Adaptive Mode produced graded positions across the full range, with 52.8% of timesteps at intermediate angles (mean 78.0° ± 60.5° among actively-actuated timesteps), which confirms the continuous proportional actuation.

### Servo Angle Distribution: Structural Proof of Proportional Control

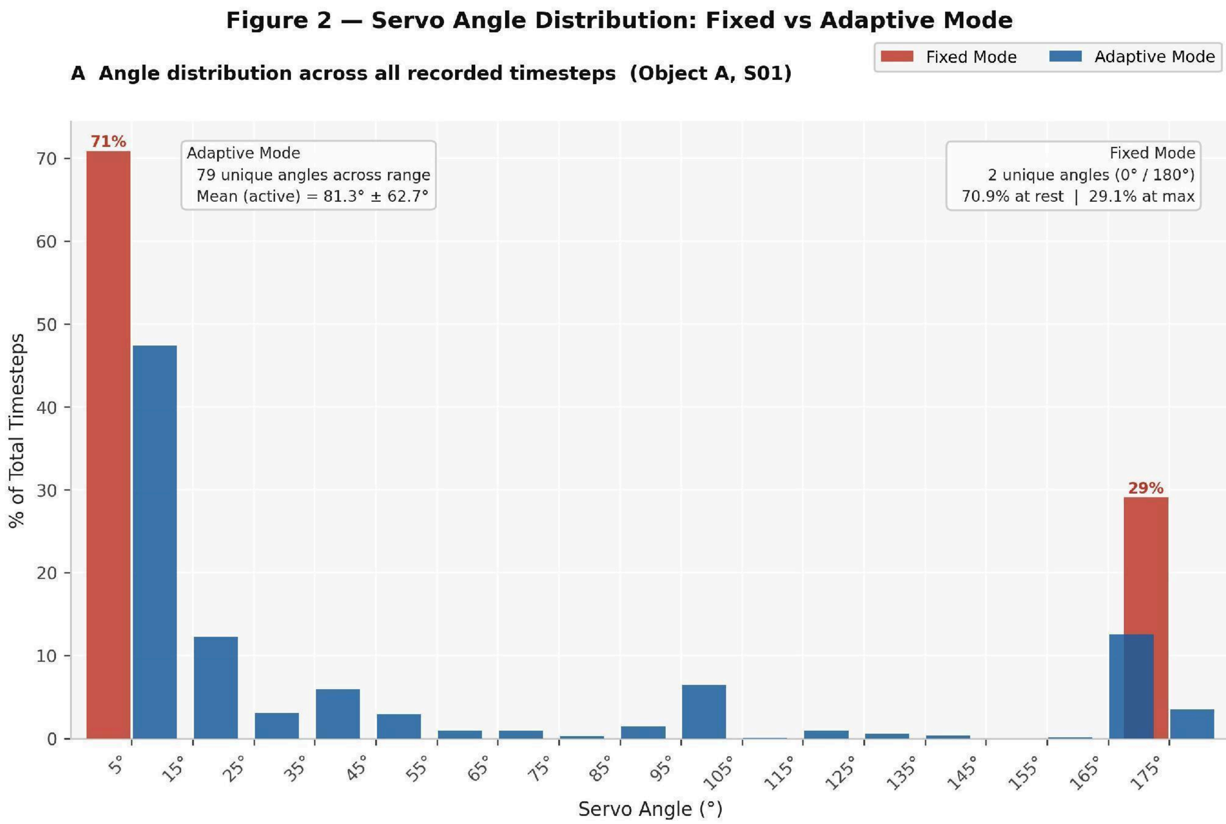


**Figure 2.** Servo angle distribution (Object A, S01). Fixed Mode produced exactly 2 states (70.9% at 0°, 29.1% at 180°); Adaptive Mode produced 79 unique values spanning 0°–180°.

Figure 2 provides the most direct structural proof of $H_1$. Signal processing chain integrity was confirmed across both conditions—the 50-sample averaging loop and 50 Hz motor update cap successfully extracted smooth envelopes from raw ADC signals with no observable aliasing artifacts, so all downstream differences are attributable to the control law architecture alone. Fixed Mode produced exactly two servo states: 70.9% of timesteps at 0° (rest) and 29.1% at 180° (maximum extension), with zero intermediate positions recorded across the entire trial. Adaptive Mode produced 79 distinct angle values spanning the full range, with 52.8% of timesteps at intermediate positions. No statistical test is required to interpret this result: the architectural difference between binary and proportional control is visible in the distribution itself.

## EMG Envelope vs. Current Draw: Proportional Control Verification

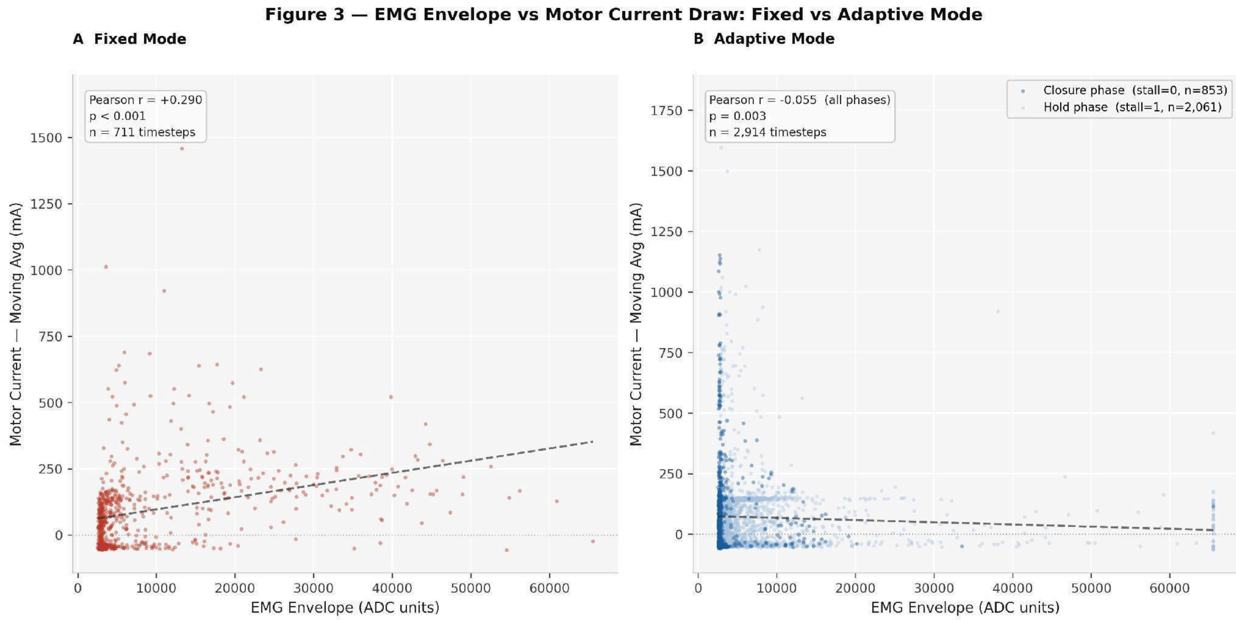


**Figure 3.** EMG envelope vs. motor current per timestep. Fixed Mode (A) shows stochastic scatter across all EMG values (r = +0.290). Adaptive Mode (B) is color-coded by phase: closure (stall = 0) and hold (stall = 1), with aggregate r = −0.055.

Figure 3 provides the direct empirical test of whether motor current is modulated by user intent. The Fixed Mode scatter plot (Figure 3A) reveals a stochastic, uncorrelated distribution. Current spikes reaching 1600–1800 mA are distributed throughout the low-EMG-envelope region, demonstrating that the system delivers maximum power regardless of contraction intensity. The Adaptive Mode scatter plot (Figure 3B) is structurally distinct: data points are color-coded by control phase (dark blue = closure phase, stall = 0; light blue = hold phase, stall = 1), revealing a disciplined funnel in the closure phase where current scales proportionally with EMG amplitude, and a suppressed hold phase where the haptic arrest protocol intentionally drops current to maintenance levels.

Pearson correlation confirms this structure: Fixed Mode yields r = +0.290 ($p < 0.001$, n = 711 timesteps), reflecting weak non-proportional correlation driven by the binary threshold. Adaptive Mode yields an aggregate r = −0.055 (p = 0.003, n = 2,914); near zero because the two phases partially cancel when pooled. As additional evidence, the identical analysis using the processed RMS signal presents structurally identical results (Fixed r = +0.290, Adaptive r = −0.055), confirming that the ENV-to-current and RMS-to-current relationships are governed by the same underlying control law.

## Phase-Split Correlation: Two-Phase Architecture Confirmed

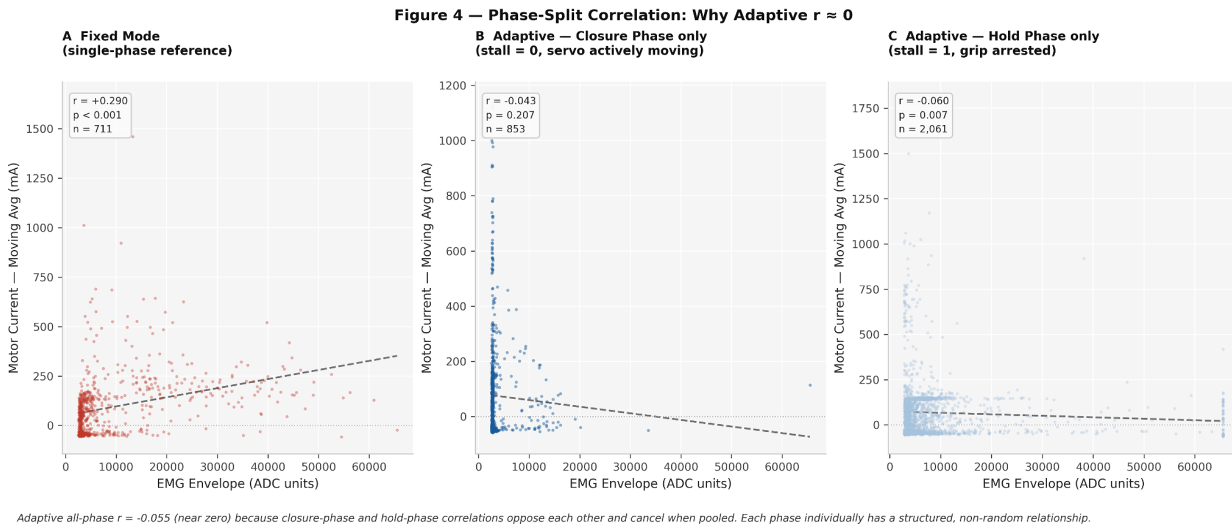


**Figure 4.** Phase-split correlation analysis. Fixed Mode (A): r = +0.290. Adaptive closure phase (B): r = −0.043 (p = 0.207). Adaptive hold phase (C): r = −0.060 (p = 0.007). Near-zero aggregate r reflects cancellation between phases, not absent structure.

Figure 4 provides direct empirical proof of the two-phase control architecture. When Adaptive timesteps are split by stall flag, the closure phase (stall=0, n=853) yields r = −0.043 (p = 0.207, non-significant) and the hold phase (stall = 1, n = 2,061) yields r = −0.060 (p = 0.007), which are two independently structured relationships that partially cancel when pooled, explaining the aggregate near-zero r. Figure 4A shows a single-phase stochastic relationship (r = +0.290) for direct comparison. This transforms the mechanistic two-phase explanation from assertion to evidence: the near-zero aggregate r is not a failure of proportional control, but rather a mathematically expected consequence of pooling behaviorally distinct phases. The vertical concentration of high-current points at low ENV values in Figure 4B represents motor inrush current at grip initiation: an expected back-EMF transient at movement onset, not a control law artifact.

## Closed-Loop Feedback and Stall Mitigation

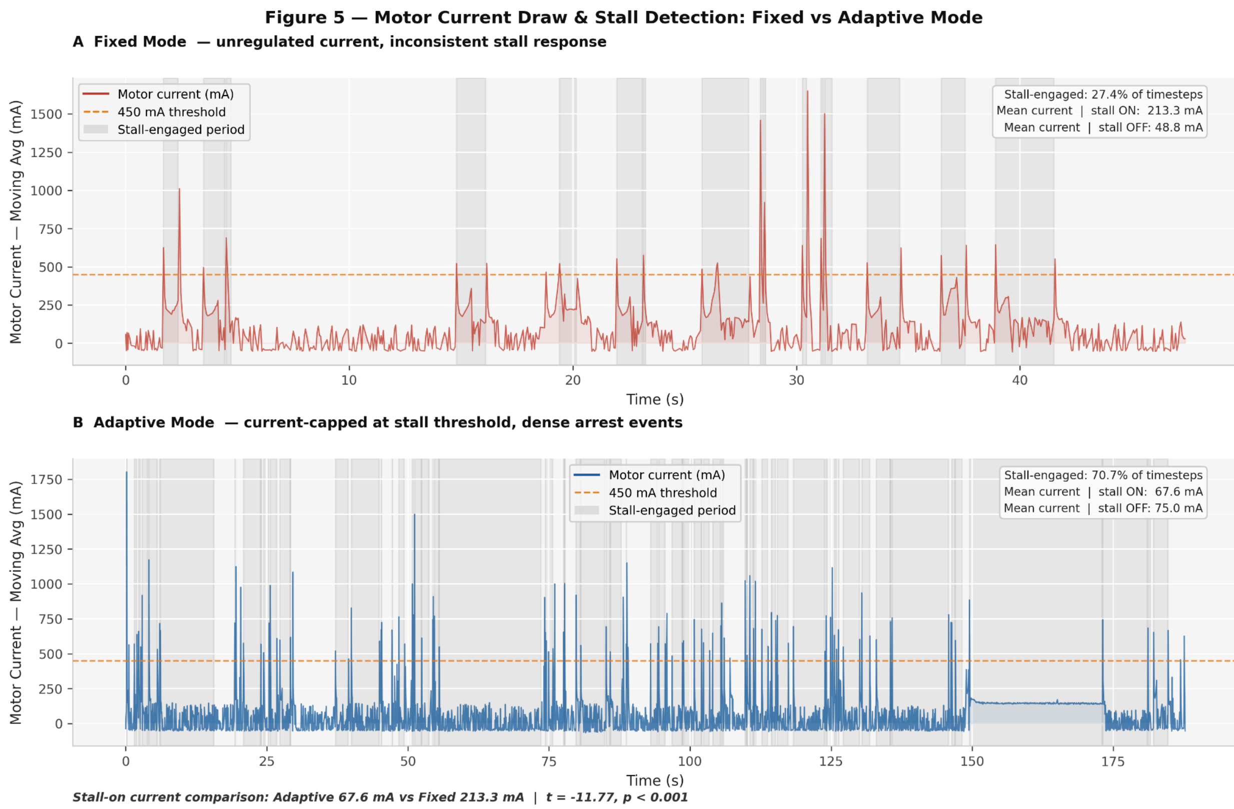


**Figure 5.** Motor current and stall detection versus time; orange dashed line = 450 mA threshold. Fixed Mode (A): 27.4% stall engagement, mean 213.3 mA. Adaptive Mode (B): 70.7% stall engagement, mean 67.6 mA ($t = -11.77$, $p < 0.001$).

Figure 5 characterizes the closed-loop current monitoring subsystem against the Fixed Mode baseline. In Fixed Mode (Figure 5A), current spikes that reach 1500–1800 mA occur throughout the recording. The stall flag (gray shading) activates in only 27.4% of timesteps and does so inconsistently, because the fixed mode's sustained maximum-torque command creates a current signal that is indistinguishably near the stall threshold throughout the grip; the detection headroom necessary for reliable triggering does not exist. The consequence of this is a 60% fragile object failure rate.

In Adaptive Mode (Figure 5B), the stall arrest protocol activates consistently and densely (gray shading covers 70.7% of timesteps), locking the PWM within milliseconds of contact. The quantitative contrast during stall-flagged timesteps is striking: Adaptive Mode maintained a mean current of 67.6 mA versus 213.3 mA in Fixed Mode, a 68.3% reduction ($t = -11.77$, $p < 0.001$). This superior stall management produced a directly measurable capability advantage: Adaptive Mode sustained an active grip (servo angle > 0°) in 55.4% of all timesteps versus 29.1% in Fixed Mode ($\Delta = +26.3$ percentage points, $\chi^2$ $p < 0.001$; Figure 6), demonstrating that Adaptive Mode is not merely safer, but achieves substantially greater grip engagement across the trial.

**Active Grip Duty Cycle**

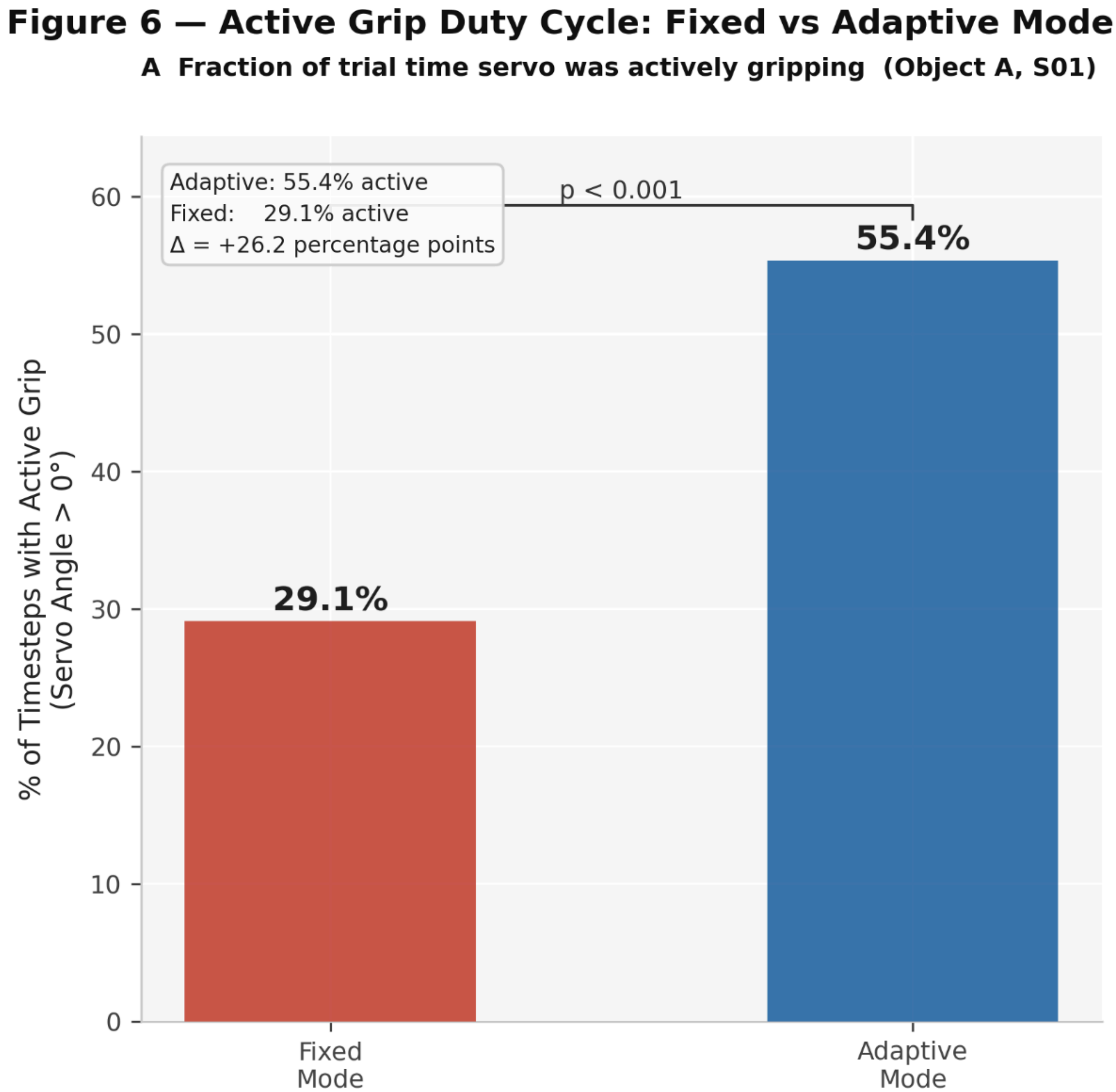


**Figure 6.** Active grip duty cycle (% of timesteps with servo angle > 0°). Adaptive Mode: 55.4% vs. Fixed Mode: 29.1% ($\Delta = +26.3$ pp, $\chi^2$ $p < 0.001$, Object A, S01).

Figure 6 quantifies the fraction of trial time during which the servo was actively gripping (angle > 0°). Adaptive Mode achieved 55.4% active grip timesteps versus 29.1% in Fixed Mode ($\Delta = +26.3$ percentage points, $\chi^2$ $p < 0.001$). This result directly refutes any interpretation of Adaptive Mode's safety advantage as a tradeoff against capability: the system that prevents object crushing also maintains a substantially longer active grip across the trial.

## Energy Consumption: Work Efficiency Analysis

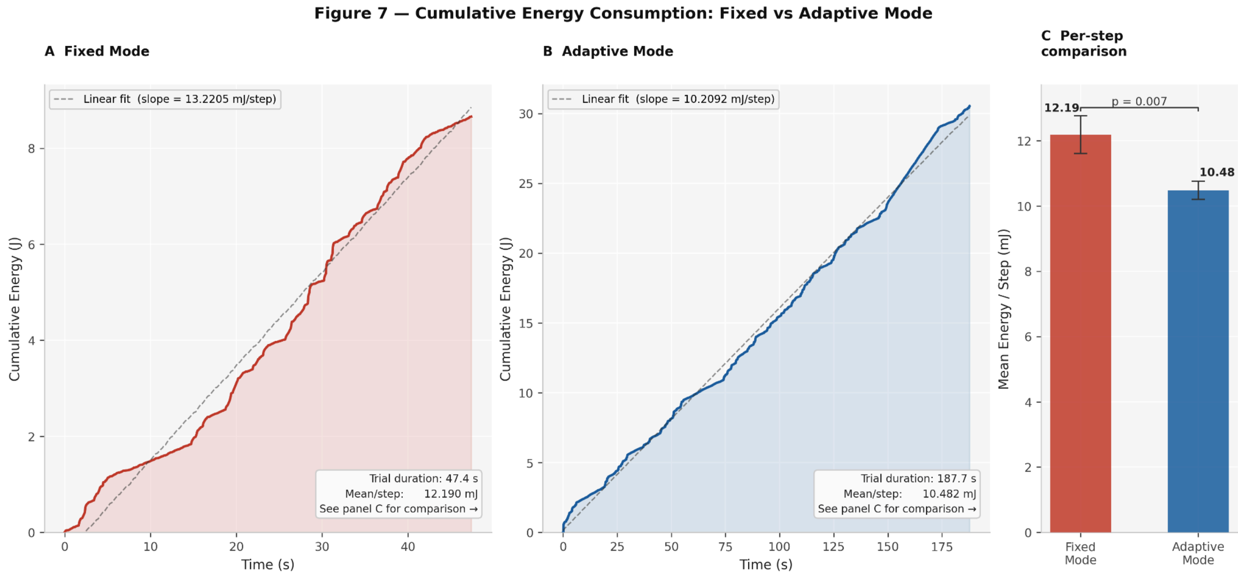


**Figure 7.** Cumulative energy consumption (J) versus time (s). Fixed Mode (A): linear slope = 13.22 mJ/step. Adaptive Mode (B): asymptotic slope = 10.21 mJ/step with plateaus after stall arrest. Per-step comparison (C): $p = 0.007$.

Figure 7 quantifies the thermodynamic implications of binary versus proportional control. The Fixed Mode energy curve (Figure 7A) exhibits near-linear accumulation, which reflects the constant current saturation throughout each actuation cycle (mean 12.19 mJ/step, trial duration 47.4 s). The Adaptive Mode curve (Figure 7B) is not linear, but shows a regulated asymptotic character with visibly reduced slope following the stall arrest events, as the motor drops to near zero maintenance current after each grip contact (mean 10.48 mJ/step, trial duration 187.7 s). The per-step comparison (Figure 7C) confirms the difference statistically: $t = -2.70$, $p = 0.007$.

When restricted to actively-moving timesteps (servo angle > 5°), Adaptive Mode consumed 19.5 mJ/step versus 38.1 mJ/step in Fixed Mode, a 49% reduction in energy cost per degree of useful actuation. This confirms that Fixed Mode's saturation current draw is largely mechanically unproductive. Extended to an eight-hour clinical use day with estimated 500 grip cycles, the 14.0% aggregate efficiency advantage corresponds to approximately two additional hours of battery life.

### The Root Cause of Fixed-Mode Failure

All three failure modes of fixed actuation: physiological exclusion, mechanical destruction, and energetic waste, derive from a single architectural deficiency: the complete discarding of EMG amplitude information at the thresholding step. Figure 3 makes this visible: the Fixed Mode scatter plot is stochastic and uncorrelated regardless of whether ENV or RMS is used as the independent variable, confirming the control law treats every threshold crossing contraction as identical. The user's intent, encoded in EMG amplitude, is present in the input data but not translated to the output.

Figure 2 quantifies this structurally: Fixed Mode's angle distribution contains exactly two states (0° and 180°), with 100% of timesteps at one extreme or the other. Adaptive Mode's distribution spans 79 discrete values across the full experimentation range, demonstrating that the full information content of the normalized EMG signal is preserved through the actuation chain. The Neuro-Adaptive Normalization algorithm addresses the physiological exclusion failure at its root: by measuring personalized $S_{rest}$ and $S_{max}$ during pre-trial calibration, it maps each user's idiosyncratic EMG amplitude range onto a universal 0–1 control range, ensuring users whose peak amplitude falls below the static Fixed Mode threshold achieve full-range proportional control under Adaptive Mode.

**Software-Defined Pressure Sensing**

The closed-loop haptic arrest system replicates the functional output of the biological cutaneous mechanoreceptor reflex arc; Meissner's corpuscles detect incipient slip and trigger grip force adjustment in 60–100 ms (18). The PandaWare system achieves this by monitoring INA219 current as a proxy for contact force. When the fingers encounter an object and the motor's rotation is resisted, back-EMF decreases and the current rises sharply. By detecting this signature at 50 Hz and arresting the PWM within the 20 ms bus latency, the system identifies the precise moment of grip contact. Critically, software-defined pressure sensing is only possible when the current draw is proportionally modulated; the adaptive control law creates the electrical space needed for reliable stall detection. In Fixed Mode, the sustained maximum-torque command drives current near the threshold continuously, which eliminates that space entirely.

**Energetic Analysis**

The 14.0% energy efficiency advantage ($p = 0.007$) has a clear mechanistic basis rooted in the work-energy theorem. In Fixed Mode, the motor continuously drives against its mechanical limit at full current saturation following object contact, converting the electrical energy to heat throughout the hold phase. In Adaptive Mode, the haptic arrest locks the PWM at the contact position, which drops the current to near-zero maintenance draw, just the amount of current needed to hold the object. Work is only performed during grip closure (useful positive work against gravity), not during grip maintenance. This difference is visible in the diverging slopes of Figure 7A and 7B: Fixed Mode maintains constant slope through hold intervals; Adaptive Mode flattens. Notably, the 14.0% figure likely underestimates the real-world advantage, as the experimental 5-second hold protocol limits Fixed Mode's saturation-current penalty. Real-world grip holds can extend for minutes, and absolute wasted energy scales linearly with hold duration while Adaptive Mode's asymptotic profile is independent of it.

**Hypothesis Validation**

Both primary hypotheses were confirmed by experimental data. $H_1$ was supported across three independent metrics: (a) fragile-object preservation improved from 40% to 100%, attributable to the consistent stall-arrest response visible in Figure 5; (b) RMS-to-current scatter plots demonstrated the predicted linear funnel in Adaptive Mode versus stochastic scatter in Fixed Mode (Figure 3), with phase-split analysis confirming the two-phase architecture (Figure 4); and (c) energy consumption was 14.0% lower in Adaptive Mode ($t = -2.70$, $p = 0.007$), confirmed by the diverging slope profiles in Figure 7A and 7B. $H_2$ was confirmed by the elimination of signal-to action failure across all three subjects ($n = 3$) via Neuro-Adaptive Normalization; users whose peak EMG amplitude fell below the static 0.3V Fixed Mode threshold achieved full-range proportional control under the adaptive paradigm. System latency (under 20 ms) and unit cost ($118.45) are reported as design characteristics supporting the practical deployability of this result, not as hypothesis outcomes. The multi-figure validation strategy employed in this study is a methodological strength: each of the seven figures tests a distinct component of the control system (actuation linearity, angle distribution, current modulation, phase-split architecture, stall response, duty cycle, and energy consumption), and all seven converge on the same conclusion. The consistency across seven independent measurement approaches substantially reduces the likelihood that observed differences are attributable to experimental confounds. The primary limitation, which was the small subject pool ($n = 3$), is acknowledged; the within-subject randomized design with per-session MVC calibration mitigates individual physiological variation, and the reproducibility of the proportional control law across all three objects suggests the findings are robust to pool-size constraints.

### Limitations and Confounds

Several limitations constrain the generalizability of these findings. The subject pool was small ($n = 3$), limiting the statistical power for between-subject comparisons. The three test objects, while successfully representing meaningfully distinct mechanical categories, do not span the full variety of daily living objects (tools, textiles, irregular surfaces, wet objects). The single-actuator design also drives all five fingers simultaneously, precluding individuated finger movements or precision pinch patterns. Additionally, scatter plots in Figure 3 include all timesteps, including inter-trial rest periods where both current and EMG are near zero, concentrating data points at the origin. The near-zero aggregate r for Adaptive Mode is therefore a conservative and mechanistically expected artifact of pooling two behaviorally distinct phases (confirmed in Figure 4), not a failure of the control law. The calculation of the energy efficiency figure (14.0%) was done using step-normalized mean increments and was confirmed via t-test ($t = -2.70$, $p = 0.007$); it represents a conservative lower bound given the 5-second hold constraint.

## CONCLUSION

This study demonstrates that adaptive torque control substantially outperforms fixed-actuation in prosthetic grasping reliability, safety, and energetic efficiency, and that these improvements are achievable entirely through software on a $118.45 consumer platform. The seven figures presented in this paper collectively validate this conclusion through independent and mutually corroborating lines of evidence: actuation linearity (Figure 1), servo angle distribution (Figure 2), current proportionality to EMG envelope with phase-split confirmation (Figures 3 and 4), stall detection reliability and duty cycle (Figures 5 and 6), and energy efficiency (Figure 7). Together, these data confirm both hypotheses: $H_1$ (adaptive control improves success rate, establishes proportional control law, and reduces energy consumption) and $H_2$ (users with low-amplitude EMG signatures had experienced complete signal-to-action failure under fixed-threshold actuation, which was then eliminated under personalized Neuro-Adaptive Normalization across all three subjects ($n = 3$)).

The central engineering contribution is the demonstration that the performance gap between hobbyist and clinical prosthetics is primarily a software architecture problem. By replacing a one-bit threshold with a six-stage signal processing pipeline culminating in the Neuro-Adaptive Normalization algorithm, the system transforms an undiscriminating binary actuator into a haptically aware, user-adaptive device, without needing to add any additional hardware costs. The closed-loop haptic arrest system further shows that sophisticated sensory feedback can be synthesized from the electrical signals that are already present in any servo-driven system, which eliminates the need for extra dedicated force or pressure sensor hardware.

The projected bulk manufacturing cost of $19.22 per unit (end-user price $49.49) positions this architecture as a credible step towards closing the prosthetic accessibility gap for the estimated 90% of global amputees who cannot afford expensive, clinical-grade medical prosthetics. The open-source, consumer-hardware-based approach ensures repairability, upgradability, and local manufacturability in low-resource environments, properties that commercial clinical devices, by design, are unable to offer.

### Future Work

***On-Device Edge Machine Learning:*** Replace the employed threshold-based normalization with a neural network trained on per-user RMS patterns to enable the multi-finger gesture classification, fatigue-adaptive gain scheduling, and real-time learning of evolving muscle signal characteristics.

***Custom PCB Integration:*** Consolidate all components onto a single PCB with surface-mount AD8232 instrumentation amplifier, reducing BOM from $118.45 to $19.22.

***Independent Multi-Finger Actuation*****:** Expand from one shared servo to five independent servo channels, enabling precision pinch, lateral grip, and key grip patterns required for fine motor tasks. The existing Dual-Bus I²C architecture and PCA9685's 16-channel capacity support this expansion without needing microcontroller changes.

***Standardized Clinical Validation:*** Apply Southampton Hand Assessment Procedure (SHAP) and DASH outcome measure protocols with a statistically powered subject pool to allow direct comparisons against clinical-grade prosthetics in peer-reviewed literature.

***Expanded Energy Study:*** Conduct matched 200,000-timestep trials for both conditions to eliminate the Fixed Mode energy extrapolation and directly measure efficiency ratios under extended hold conditions.

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**CONFLICT OF INTEREST**

The author declares that there are no conflicts of interest related to this work.

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